

Periodic classification of ELEMENTS

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1. Hydrogen (H)	Hi	11. Sodium (Na)	Na
2. Helium (He)	Hello	12. Magnesium (Mg)	Mango
3. Lithium (Li)	Listen	13. Aluminium (Al)	Allu
4. Beryllium (Be)	Be	14. Silicon (Si)	Se
5. Boron (B)	B	15. Phosphorus (P)	Pepsi
6. Carbon (C)	C	16. Sulphur (S)	Soda
7. Nitrogen (N)	News	17. Chlorine (Cl)	Cola
8. Oxygen (O)	On	18. Argon (Ar)	Aur
9. Fluorine (F)	Friday	19. Potassium (K)	Kaju
10. Neon (Ne)	Night	20. Calcium (Ca)	catli

s - block

Group 1

Hydrogen (H)	H
Lithium (Li)	Li
Sodium (Na)	Na
Potassium (K)	K
Rubidium (Rb)	Rb
Cesium (Cs)	Cs
Francium (Fr)	Farar

Group 2

Beryllium (Be)	Beta
Magnesium (Mg)	Mange
Calcium (Ca)	Car
Strontium (Sr)	Scooter
Barium (Ba)	Baap
Radium (Ra)	Razi

p - block

					18
13	14	15	16	17	He
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	Rn

13. B Al Gaya Insaan Takla

14. Come See Girls in Sunday Pub

15. NZ Pakistan Australia Sab Bhikari

16. O S Se Tel Po

17. First Class Biryani I Ate

18. He Narayan Arjun Kare Xerox Ramayan

1. Rearrange the columns 2 and 3 so as to match with the column 1.

Column 1	Column 2	Column 3
Triad	Average of the first and the third atomic mass	Dobereiner
Octave	Properties of the eighth element similar to the first	Newlands
Atomic Number	Positive charge on the nucleus	Moseley
Period	Sequential change in molecular formulae	Mendeleev
Nucleus	Concentrated mass and positive charge	Rutherford
Electron	Lightest and negatively charged particle in all the atoms	Thomson

2. Choose the correct option and rewrite the statement.

- a. The number of electrons in the outermost shell of alkali metals is 1.
(i) 1 (ii) 2 (iii) 3 (iv) 7
- b. Alkaline earth metals have valency 2. This means that their position in the modern periodic table is in Group 2.
(i) Group 2 (ii) Group 16 (iii) Period 2 (iv) d-block

c. Molecular formula of the chloride of an element X is XCl . This compound is a solid having high melting point. Which of the following elements be present in the same group as X.

(i) Na (ii) Mg (iii) Al (iv) Si

d. In which block of the modern periodic table are the nonmetals found?

(i) s-block (ii) p-block (iii) d-block (iv) f-block

3. An element has its electron configuration as 2,8,2. Now answer the following questions.

a. What is the atomic number of this element?

Electronic configuration of this element is (2,8,2)

$\therefore$ The atomic number of this element is 12.

b. What is the group of this element?

Electronic configuration is (2,8,2). As it has 2 valence electrons, this element belongs to group 2 and its name is Magnesium (Mg).

c. To which period does this element belong?

Electronic configuration is (2,8,2). It has 3 shells. The number of shells indicates the period number. Hence it belongs to period 3.

d. With which of the following elements would this element resemble?

(Atomic numbers are given in the brackets)

N (7), Be (4) , Ar (18), Cl (17)

This element resembles Be (4) whose electronic configuration is (2,2). Both these elements have 2 valence electrons and they belong to group 2.

4. Write down the electronic configuration of the following elements from the given atomic numbers. Answer the following question with explanation.

a. ${}_3\text{Li}$, ${}_{14}\text{Si}$, ${}_2\text{He}$, ${}_{11}\text{Na}$, ${}_{15}\text{P}$

Which of these elements belong to be period 3?

Electronic configuration of the given elements is:

$${}_3\text{Li} = 2,1$$

$${}_{14}\text{Si} = 2,8,4$$

$${}_2\text{He} = 2$$

$${}_{11}\text{Na} = 2,8,1$$

$${}_{15}\text{P} = 2,8,5$$

${}_{14}\text{Si}$, ${}_{11}\text{Na}$, ${}_{15}\text{P}$ belong to the third period because according to their electronic configuration, each element contains three shell i.e. K,L,M.

b. ${}_1\text{H}$, ${}_7\text{N}$, ${}_{20}\text{Ca}$, ${}_{16}\text{S}$, ${}_4\text{Be}$, ${}_{18}\text{Ar}$

Which of these elements belong to the second group?

Electronic configuration of the given elements is:

$${}_1\text{H} = 1$$

$${}_7\text{N} = 2,5$$

$${}_{20}\text{Ca} = 2,8,8,2$$

$${}_{16}\text{S} = 2,8,6$$

$${}_4\text{Be} = 2,2$$

$${}_{18}\text{Ar} = 2,8,8$$



${}_{20}\text{Ca}$, ${}_4\text{Be}$ belong to second group because these elements have 2 electrons in their outermost shell.

c. ${}_7\text{N}$, ${}_6\text{C}$, ${}_8\text{O}$, ${}_5\text{B}$, ${}_{13}\text{Al}$.

Which is the most electronegative element among these?

Electronic configuration of the given elements is:

$${}_7\text{N} = 2, 5$$

$${}_6\text{C} = 2, 4$$

$${}_8\text{O} = 2, 6$$

$${}_5\text{B} = 2, 3$$

$${}_{13}\text{Al} = 2, 8, 3$$

${}_8\text{O}$ is the most electronegative element among these because electronegativity increases as we move from left to right in a period of the periodic table.

d. ${}_4\text{Be}$, ${}_6\text{C}$, ${}_8\text{O}$, ${}_5\text{B}$, ${}_{13}\text{Al}$

Which is the most electropositive element among these?

Electronic configuration of the given elements is:

$${}_4\text{Be} = 2, 2$$

$${}_6\text{C} = 2, 4$$

$${}_8\text{O} = 2, 6$$

$${}_5\text{B} = 2, 3$$

$${}_{13}\text{Al} = 2, 8, 3$$

${}_4\text{Be}$ holds the highest electropositivity among the given elements due to its strong metallic character resulting from its position on the left side of the periodic table. This positioning grants it a low ionization energy and a greater tendency to lose electrons, showcasing a higher electropositive nature compared to the other elements like ${}_6\text{C}$, ${}_8\text{O}$, ${}_5\text{B}$, and ${}_{11}\text{Al}$.

e. ${}_{11}\text{Na}$, ${}_{15}\text{P}$, ${}_{17}\text{Cl}$, ${}_{14}\text{Si}$, ${}_{12}\text{Mg}$

Which of these has largest atoms?

Electronic configuration of the given elements is:

${}_{11}\text{Na} = 2, 8, 1$

${}_{15}\text{P} = 2, 8, 5$

${}_{17}\text{Cl} = 2, 8, 7$

${}_{14}\text{Si} = 2, 8, 4$

${}_{12}\text{Mg} = 2, 8, 2$

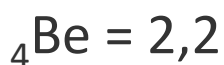
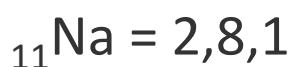
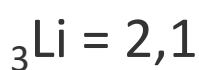
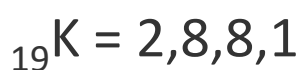
Among these elements, ${}_{11}\text{Na}$ has the largest size. This is because, in a period, the atomic radius gets smaller from left to right. As the atomic number increases, the number of protons and electrons also increase in the atoms. The strong positive charge in the nucleus pulls the electrons closer to it, causing the atom's size to decrease.

f. ${}_{19}\text{K}$, ${}_3\text{Li}$, ${}_{11}\text{Na}$, ${}_4\text{Be}$

Which of these atoms has smallest atomic radius?



Electronic configuration of the given elements is:

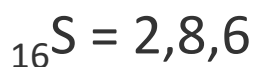
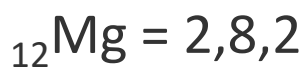
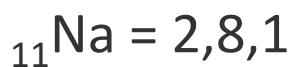
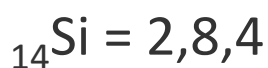
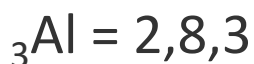


${}_4\text{Be}$ has smallest atomic radius because ${}_{19}\text{K}$, ${}_3\text{Li}$, ${}_{11}\text{Na}$ are present in same group 1 but Be is present in group 2. According to the trend, the atomic radius gets smaller from left to right. As the atomic number increases, the number of protons and electrons also increase in the atoms. The strong positive charge in the nucleus pulls the electrons closer to it, causing the atom's size to decrease.

g. ${}_{13}\text{Al}$, ${}_{14}\text{Si}$, ${}_{11}\text{Na}$, ${}_{12}\text{Mg}$, ${}_{16}\text{S}$

Which of the above elements has the highest metallic character?

Electronic configuration of the given elements is:



${}_{11}\text{Na}$ has the highest metallic character because metallic character of an element decreases as we move from left to right in the modern periodic table. This happens as the tendency of an atom to lose electrons decreases due to gradual increase in the number of protons and nuclear charge.

h. ${}_6\text{C}$, ${}_3\text{Li}$, ${}_9\text{F}$, ${}_7\text{N}$, ${}_8\text{O}$

Which of the above elements has the highest non-metallic character?

Electronic configuration of the given elements is:

$${}_6\text{C} = 2,4$$

$${}_3\text{Li} = 2,1$$

$${}_9\text{F} = 2,7$$

$${}_7\text{N} = 2,5$$

$${}_8\text{O} = 2,6$$

${}_9\text{F}$ has the highest non-metallic character because non-metallic character of an element increases as we move from left to right in a period. This happens as the tendency of an atom to gain electrons increases due to increase in nuclear charge, the valence electrons are pulled in strongly by the nucleus and it becomes easier for an atom to gain electrons.

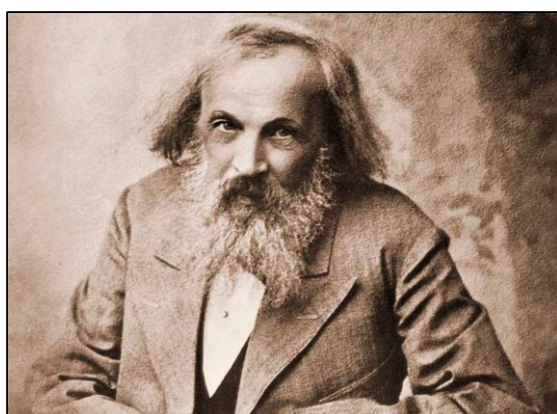
5. Write the name and symbol of the element from the description.

- a. **The atom having the smallest size** - Helium (He).
- b. **The atom having the smallest atomic mass** - Hydrogen (H).
- c. **The most electronegative atom** - Fluorine (F).
- d. **The noble gas with the smallest atomic radius** - Helium (He).
- e. **The most reactive nonmetal** - Fluorine (F).

6. Write short notes.

a. Mendeleev's Periodic law

- i. The Russian scientist Dmitri Mendeleev developed the periodic table of elements during the period 1869 to 1872 A.D.
- ii. Mendeleev's periodic table is the most important step in the classification of elements.
- iii. Mendeleev considered the fundamental property of elements, namely the atomic mass as standard and arranged 63 elements known at that time in an increasing order of their atomic masses.
- iv. Then, he transferred this into the periodic table of elements in accordance with the physical and chemical properties of these elements.
- v. He organized the periodic table on the basis of the chemical and physical properties of the elements.
- vi. These were the molecular formulae of hydrides and oxides of the elements, melting points, boiling points and densities of the elements and their hydrides and oxides.
- vii. Mendeleev found that the elements with similar physical and chemical properties repeat after a definite interval.
- viii. On the basis of this finding, Mendeleev, stated the following periodic law "Properties of elements are periodic function of their atomic masses".



b. Structure of the Modern Periodic table.

	1																18	
1	1	2															2	
2	3	4															10	
3	11	12	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
4	19	s					d								p			
5																		
6			57-71															
7			89-103															118
Lanthanides			57 La	f													71	
Actinides			89 Ac														103	

Image just for reference. Not to be drawn in exam

- The Modern Periodic table contains seven horizontal rows called the periods 1 to 7.
- Similarly, eighteen vertical columns are the groups 1 to 18.
- The arrangement of the periods and groups results into formation of boxes.
- Atomic numbers are serially indicated in the upper part of these boxes.
- Each box corresponds to the place for one element.
- Apart from these seven rows, two separate rows are placed at the bottom of the periodic table.
- These are called Lanthanide series and Actinide series respectively.
- There are 118 boxes in the periodic table including the two series.
- It means that there are 118 places for elements in the Modern Periodic table.

- x. The Modern Periodic table is now completely filled, means all the 118 elements are now discovered.
- xi. The entire periodic table is divided into four blocks, viz, s-block, p-block, d-block and f-block.
- xii. The s-block contains the groups 1 and 2 elements.
- xiii. The groups 13 to 18 constitute the p-block elements.
- xiv. The groups 3 to 12 constitute the d-block elements. The d-block elements are called transition elements.
- xv. The lanthanide and actinide series at the bottom form the f-block elements.
- xvi. A zig-zag line can be drawn in the p-block of the periodic table.
- xvii. The metalloid elements lie along the border of the zig-zag line.
- xviii. All the metals lie on the left side of the zig-zag line, while all the non-metals lie on the right side.

c. Position of isotopes in the Mendeleev's and the Modern Periodic table.

- i. Different atoms of the same element having same atomic number, but different atomic mass number are called isotopes.
- ii. Isotopes were discovered long time after Mendeleev put forth the periodic table.
- iii. As isotopes have the same chemical properties but different atomic masses, a challenge was posed in placing them in Mendeleev's periodic table.

- iv. Based on chemical properties, they were to be placed in same group, while based on atomic masses they were to be placed in different groups.
- v. Hence, there was no definite place for isotopes.
- vi. This anomaly arose in Mendeleev's periodic table because the elements were arranged in an increasing order of their atomic masses.
- vii. In 1913 A.D., Henry Moseley revealed that atomic number is a more fundamental property of an element than its atomic mass.
- viii. Hence, he modified Mendeleev's Periodic Table and arranged the elements in an increasing order of their atomic number in the Modern Periodic Table.
- ix. By doing so, the anomaly of Mendeleev's Periodic Table regarding isotopes was removed.
- x. Therefore, in the Modern Periodic Table the isotopes could be placed in the same group.

7. Write scientific reasons.

- a. **Atomic radius goes on decreasing while going from left to right in a period.**
 - i. Atomic radius is the distance between the nucleus of the atom and its outermost shell.
 - ii. atomic radius goes on decreasing while going from left to right within a period.

- iii. This is because while going from left to right within a period, the atomic number increases one by one, meaning the positive charge on the nucleus increases by one unit at a time.
- iv. However, the additional electron gets added to the same outermost shell.
- v. Due to the increased nuclear charge the electrons are pulled towards the nucleus to a greater extent and thereby the size of the atom decreases.

b. Metallic character goes on decreasing while going from left to right in a period.

- i. Metallic character is the tendency of an atom to lose electrons.
- ii. While going from left to right within a period the outermost shell remains the same.
- iii. However, the positive charge on the nucleus goes on increasing while the atomic radius goes on decreasing and thus the effective nuclear charge goes on increasing.
- iv. Due to this, the tendency of atom to lose valence electrons decreases within a period from left to right.
- v. Thus, metallic character goes on decreasing while going from left to right in a period.



c. Atomic radius goes on increasing down a group.

- i. While going down a group the atomic size goes on increasing.
- ii. This is because while going down a group a new shell is added. Therefore, the distance between the outermost electron and the nucleus goes on increasing.
- iii. As a result of this the atomic size increases despite the increased nuclear charge.

d. Elements belonging to the same group have the same valency.

- i. Valency is the number of electrons donated, accepted or shared by the atoms of an element to attain noble gas configuration.
- ii. Valency is dependent upon the number of electrons present in the outermost shell of an atom called valence electrons.
- iii. Elements in the same group have same number of valence electrons, irrespective of the number of shells.
- iv. Hence, elements in the same group have the same valency.

e. The third period contains only eight elements even though the electron capacity of the third shell is 18.

- i. The third period contains only eight elements because the Aufbau principle dictates that electrons fill lower-energy orbitals before higher-energy ones.
- ii. In the third period, after filling the 3s and 3p orbitals, the next available 3d orbital is at slightly higher energy than the 4s orbital.

- iii. Therefore, the 4s orbital is filled before the 3d orbital, resulting in the completion of the period with eight elements.

8. Write the names from the description.

- a. The period with electrons in the shells K, L and M.

Period 3

- b. The group with valency zero.

Group 18

- c. The family of nonmetals having valency one.

Halogen family

- d. The family of metals having valency one.

Alkali metals

- e. The family of metals having valency two.

Alkaline earth metals

- f. The metalloids in the second and third periods.

Boron and Silicon

- g. Nonmetals in the third period.

Phosphorus, Sulphur, Chlorine and Argon

- h. Two elements having valency 4.

Carbon and Silicon



INTEXT QUESTIONS

1. What are the types of matter?

There are three types of matters viz elements, compounds and mixtures.

2. What are the types of elements?

The three types of elements are metals, non-metals and metalloids.

3. What are the smallest particles of matter called?

The smallest particles of matter are called atoms (element) and molecules (compound).

4. What is the difference between the molecules of elements and compounds?

Molecules of elements contain atoms of the same element where as molecules of compounds contain atoms of two or more different elements.

5. Identify Dobereiner's triads from the following groups of elements having similar chemical properties.

i. Mg (24.3), Ca (40.1), Sr (87.6)

Therefore, mean of $\underline{a} + \underline{c} = \underline{111.90} = 55.95$

which is not approximately equal to the atomic mass of middle element Ca. Therefore, it does not form a Dobereiner's triad.

ii. S (32.1), Se (79.0), Te (127.6)

$$\text{Therefore, mean of } \frac{a + c}{2} = \frac{159.70}{2} = 79.85$$

which is approximately equal to atomic mass of Se. Hence, it forms a Dobereiner's triad.

iii. Be (9.0), Mg (24.3), Ca (40.1)

$$\text{Therefore, mean of } \frac{a + c}{2} = \frac{49.10}{2} = 24.55$$

which is approximately equal to atomic mass of Mg. Hence, it forms a Dobereiner's triad.

6. There are some vacant places in the Mendeleev's periodic table. In some of these places the atomic masses are seen to be predicted. Enlist three of these predicted atomic masses along with their group and period.

Predicted atomic mass	Group	Period	Predicted Name	Present Names
44	III	4	Eka boron	Scandium
68	III	5	Eka aluminum	Gallium
72	IV	5	Eka silicon	Germanium

7. Chlorine has two isotopes, viz, Cl-35 and Cl-37. Their atomic masses are 35 and 37 respectively. Their chemical properties are same. Where should these be placed in Mendeleev's periodic table? In different places or in the same place?

Mendeleev arranged the elements in increasing order of their atomic masses. Since the atomic masses of the isotopes of chlorine are 35 and 37, they should be kept in different positions in the Mendeleev Periodic table.

8. Write the molecular formulae of oxides of the following elements by referring to the Mendeleev's periodic table. Na, Si, Ca, C, Rb, P, Ba, Cl, Sn.

- Na - Na_2O
- Ca - CaO
- Rb - Rb_2O
- Ba - BaO
- Sn - SnO_2
- Si - SiO_2
- C - CO_2
- P - P_2O_5
- Cl - Cl_2O_7

9. Write the molecular formulae of the compounds of the following elements with hydrogen by referring to the Mendeleev's periodic table. C, S, Br, As, F, O, N, Cl

Elements	Compound formed with H	Elements	Compound formed with H
C	CH_4	S	H_2S
Br	HBr	As	AsH_3
F	HF	O	H_2O
N	NH_3	Cl	HCl

10. How is the problem regarding the position of cobalt (^{59}Co) and nickel (^{59}Ni) in Mendeleev's periodic table resolved in modern periodic table?

- i. According to the modern periodic law, elements are arranged in the increasing order of their atomic numbers.
- ii. The atomic numbers of cobalt and nickel are 27 and 28 respectively.
- iii. So, cobalt with a lower atomic number should come before nickel.

11. How did the position of ^{35}Cl and ^{37}Cl get fixed in the modern periodic table?

- i. Modern periodic table is based on atomic number and not on atomic mass.
- ii. This is why both the isotopes of Cl having the same atomic number 17 occupy the same position in modern periodic table.

12. Can there be an element with atomic mass 53 or 54 in between the two elements, chromium ^{52}Cr and manganese ^{55}Mn ?

- i. Chromium having atomic number 24 and manganese having atomic number 25 differ from each other by 1 unit.
- ii. Hence, there can be no element with atomic mass 53 or 54 in between the two elements.

13. What do you think? Should hydrogen be placed in the group 17 of halogens or group 1 of alkali metals in the modern periodic table?

- i. The position of hydrogen in the periodic table is unclear because it resembles both the alkali metals and halogens in some of its properties.
- ii. However, it is placed in the first group, above all the alkali metals in the modern periodic table because its electronic configuration is similar to that of alkali metals.

14. Go through the modern periodic table and write the names one below the other of the elements of group 1.

H - Hydrogen

Li - Lithium

Na - Sodium

K - Potassium

Rb - Rubidium

Cs - Caesium

Fr - Francium

are the elements of group 1.

Write the electronic configuration of the first four elements in this group.

${}_1\text{H} - (1)$

${}_3\text{Li} - (2,1)$

${}_{11}\text{Na} - (2,8,1)$

${}_{19}\text{K} - (2,8,8,1)$

Which similarity do you find in their configuration?

The outermost shell contains one electron in all these elements.

How many valence electrons are there in each of these elements?

Number of valence electrons in each of these elements is 1.

15. On going through the modern periodic table, it is seen that the elements Li, Be, B, C, N, O, F and Ne belong to the period-2. Write down electronic configuration of all of them.

${}_3\text{Li} - (2,1)$

${}_4\text{Be} - (2,2)$

${}_5\text{B} - (2,3)$

${}_6\text{C} - (2,4)$

${}_7\text{N} - (2,5)$

${}_8\text{O} - (2,6)$

${}_9\text{F} - (2,7)$

${}_{10}\text{Ne} - (2,8)$

Is the number of valence electrons same for all these elements?

No, they are not the same.

Is the number of shells the same in these ?

Yes, the number of shells are the same. Each of these elements have 2 shells.

16. What are the values of 'n' for the shells K, L and M?

K – 1 L – 2 M – 3

17. What is the maximum number of electrons that can be accommodated in a shell? Write the formula.

Shell	n	$2n^2$	Electron Capacity
K	1	2×1^2	2
L	2	2×2^2	8
M	3	2×3^2	18
N	4	2×4^2	32

18. Deduce the maximum electron capacity of the shells K, L and M.

K - 2 ($2n^2 = 2 \times 1^2 = 2$)

L - 8 ($2n^2 = 2 \times 2^2 = 2 \times 4 = 8$)

M - 18 ($2n^2 = 2 \times 3^2 = 2 \times 9 = 18$)

19. What is the relationship between the electronic configuration of an element and its valency?

- From electronic configuration, we come to know the number of valence electrons, i.e. the number of electrons in the outermost shell.
- The valency of an element is determined from valence electrons.
- That is, how many electrons it has to lose, gain or share to attain noble gas configuration.

20. The atomic number of beryllium is 4 while that of oxygen is 8. Write down the electronic configuration of the two and deduce their valency from the same.

- i. Atomic number of Beryllium – 4
Electronic configuration – (2, 2)
- ii. Atomic number of Oxygen – 8
Electronic configuration – (2, 6)
- iii. Valency of Beryllium is 2,
Valency of Oxygen is, $8 - 6 = 2$

21. The table on the next page is made on the basis of the modern periodic table. Write in it the electronic configuration of the first 20 elements below the symbol and write the valency.

Symbol		Electronic configuration		Valency			
¹⁹ K		2, 8, 8, 1		1			
¹ H							² He
1							2
1							0
³ Li	⁴ Be	⁵ B	⁶ C	⁷ N	⁸ O	⁹ F	¹⁰ Ne
2,1	2,2	2,3	2,4	2,5	2,6	2,7	2,8
1	2	3	4	3	2	1	0
¹¹ Na	¹² Mg	¹³ Al	¹⁴ Si	¹⁵ P	¹⁶ S	¹⁷ Cl	¹⁸ Ar
2,8,1	2,8,2	2,8,3	2,8,4	2,8,5	2,8,6	2,8,7	2,8,8
1	2	3	4	3	2	1	0
¹⁹ K	²⁰ Ca						
2,8,8,1	2,8,8,2						
1	2						

22. What is the periodic trend in the variation of valency while going from left to right within a period? Explain your answer with reference to period 2 and period 3.

Period 2

Elements	Li	Be	B	C	N	O	F	Ne
Valency	1	2	3	4	3	2	1	0

Period 3

Elements	Na	Mg	Al	Si	P	S	Cl	Ar
Valency	1	2	3	4	3	2	1	0

For periods 2 and 3, on going from left to right, the valency increases from 1 to 4, then starts decreasing up to 0.

23. What is the periodic trend in the variation of valency while going down a group? Explain your answer with reference to the group 1, group 2 and group 18.

- All the elements in a group have the same number of valence electrons. Therefore, down the group, valency remains the same.
- Group 1 elements have 1 valence electron. Thus, valency of group 1 elements is 1.
- Group 2 elements have 2 valence electrons. Thus, valency of group 2 elements is 2.
- The valency of group 18 elements is zero because they are noble gases with completed octet in outermost shell.

24. By referring to the modern periodic table find out the period to which the above elements belong.

Element	:	O	B	C	N	Be	Li
Atomic radius (pm)	:	66	88	77	74	111	152

Elements Li, Be, B, C, N, O belong to period 2

Arrange the above elements in a decreasing order of their atomic radii.

The above elements in the decreasing order of their atomic radii are as follows:

Elements	Li	Be	B	C	N	O
Atomic radii	152	111	88	77	74	66

Does this arrangement match with the pattern of the second period of the modern periodic table?

Yes. As we move from left to right within a period the atomic number increases one by one, meaning the positive charge on the nucleus increases by one unit at a time, but the electrons are added to the same shell, thereby, increasing the pull towards the nucleus which decreases the size of the atom.

Which of the above elements have the biggest and the smallest atom?

Lithium has the biggest atom and Oxygen has the smallest atom.

What is the periodic trend observed in the variation of atomic radius while going from left to right within a period?

Atomic radius decreases from left to right within a period.

25. By referring to the modern periodic table find out the groups to which the elements belong.

Element	:	K	Na	Rb	Cs	Li
Atomic radius (pm):		231	186	244	262	152

Above elements belong to group 1.

Arrange the above elements vertically downwards in an increasing order of atomic radii.

The above elements vertically downwards in an increasing order of atomic radii are as follows:

Li	152
Na	186
K	231
Rb	244
Cs	262

Does this arrangement match with the pattern of the group 1 of the modern periodic table?

Yes. As we go down a group number of shells increases, therefore the atomic radii also increases.

Which of the above elements have the biggest and the smallest atom?

Cesium has the biggest atom and Lithium has the smallest atom.

What is the periodic trend observed in the variation of atomic radii down a group?

Atomic radii increases down a group, as number of shells are added one at a time.

26. Look at the elements of the third period. Classify them into metals and nonmetals.

The elements in the third period are Sodium (Na), Magnesium (Mg), Aluminium (Al), Silicon (Si), phosphorus (P), Sulphur (S), Chlorine (Cl) and Argon (Ar). Out of these elements, Sodium (Na), Magnesium (Mg) and Aluminium (Al) which are to the left of the periodic table are metals and phosphorus (P), Sulphur (S) and Chlorine (Cl) which are to the right of the periodic table are nonmetals.

On which side of the period are the metals? Left or right?

Metals are present on the left side of the period.

On which side of the period did you find the nonmetals?

Nonmetals are present on the right side of the period.

27. What is the cause of nonmetallic character of elements?

- i. The two factors namely, the increasing nuclear charge and decreasing atomic radius as we go from left to right within a period, are responsible for increasing the effective nuclear charge.
- ii. Therefore, the valence electrons are held with greater and greater attractive force. This is called electronegativity of an atom.
- iii. Due to increasing electronegativity from left to right within a period, the ability of an atom to become anion by accepting outside electrons goes on increasing.
- iv. The tendency of an element to form anion or the electro negativity is the nonmetallic character of an element.

28. What is the expected trend in the variation of nonmetallic character of elements from left to right in a period?

Non-metallic character increases from left to right in a period.

29. What would be the expected trend in the variation of nonmetallic character of elements down a group?

Non-metallic character decreases as we move down a group.

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Previous Year Questions

1. Write merits of Mendeleev's periodic table.

- i. Atomic masses of some elements were revised so as to give them proper place in the periodic table in accordance with their properties. For example, the previously determined atomic mass of beryllium, 14.09, was changed to the correct value 9.4, and beryllium was placed before boron.
- ii. Mendeleev kept vacant spaces in the periodic table for elements not discovered till then. Three of these unknown elements were given the names eka-boron, eka-aluminium and eka-silicon from their known neighbours and their atomic masses were indicated as 44, 68 and 72, respectively. Not only this but their properties were also predicted. Later, these elements were discovered and named as scandium (Sc), gallium (Ga) and germanium (Ge) respectively. The properties of these elements matched well with those predicted by Mendeleev.
- iii. There was no place reserved for noble gases in Mendeleev's original periodic table. However, when noble gases such as helium, neon and argon were discovered towards the end of the nineteenth century, Mendeleev created the 'zero' group without disturbing the original periodic table in which the noble gases were fitted very well.

2. Taking into consideration the period of the elements given below, answer the following questions: **Intext Question 24**

Elements	Atomic Radius (pm)
O	66
B	88
C	77
N	74
Be	111
Li	152

- Arrange the above elements in a decreasing order of their atomic radii.
- State the period to which the above elements belong.
- Why this arrangement of elements is similar to the above period of modern periodic table?
- Which of the above elements have the biggest and the smallest atom?
- What is the periodic trend observed in the variation of atomic radius while going from left to right within a period?

3. Observe the following diagram and write the answers of the following questions:

	1	2		13	14	15	16	17	18
1	H 1								He 2
2	Li 2, 1	Be 2, 2		B 2, 3	C 2, 4	N 2, 5	O 2, 6	F 2, 7	Ne 2, 8
3	Na 2, 8, 1	Mg 2, 8, 2		Al 2, 8, 3	Si 2, 8, 4	P 2, 8, 5	S 2, 8, 6	Cl 2, 8, 7	Ar 2, 8, 8
4	K 2, 8, 8, 1	Ca 2, 8, 8, 2							
5		Sr							
6		Ba							
7		Ra							

Potassium atom

Argon atom

- a. Write the atomic numbers of first two elements in the second group.

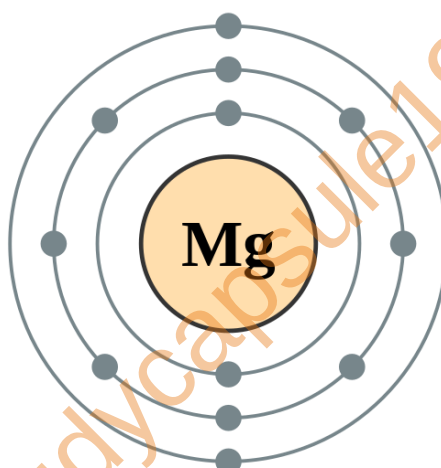
Beryllium (Be) – 4

Magnesium (Mg) - 12

- b. Write the number of valence electrons of the elements in the halogen group.

The number of valence electrons of the elements in halogen group is 7.

- c. Draw the diagram of electronic configuration of magnesium atom.



- d. After completion of a period, what change does take place in the electronic configuration of the next element?

After completion of a period a new shell is added to the next element.

- e. Write the names of any two elements from the diagram which do not take part in chemical reaction.

He, Ne , Ar

4. Elements belonging to the same group have the same valency. –
Exercise Question 7 - d

5. An element has its electronic configuration as 2, 8, 2. Now answer the following questions: **Exercise Question 3**
- What is the atomic number of this element?
 - What is the group of this element?
 - To which period does this element belong?
6. Write the modern periodic law and also give the names of 'blocks' in modern periodic table.

Modern periodic law is stated as follows:

Properties of elements are a periodic function of their atomic numbers.

The entire periodic table is divided into four blocks, viz, s-block, p-block, d-block and f-block.

7. Describe the demerits of Mendeleev's periodic table.

- The whole number atomic mass of the elements cobalt (Co) and nickel (Ni) is the same. Therefore, there was an ambiguity regarding their sequence in Mendeleev's periodic table.
- Isotopes were discovered long time after Mendeleev put forth the periodic table. As isotopes have the same chemical properties but different atomic masses, a challenge was posed in placing them in Mendeleev's periodic table.
- When elements are arranged in an increasing order of atomic masses, the rise in atomic mass does not appear to be uniform. It was not possible, therefore, to predict how many elements could be discovered between two heavy elements.

iv. No fixed position could be given to hydrogen which resembles alkali metals and halogens in the periodic table.

8. Atomic radius goes on increasing down a group. Exercise Question 7 - c

9. Define periods and groups of modern periodic table.

i. The Horizontal rows of elements in the periodic table are called periods.
The modern periodic table contains seven periods 1 to 7.

ii. The Vertical columns of elements in the periodic table are called groups.
The eighteen vertical columns in this table are the groups 1 to 18.

10. An element has its electron configuration as 2, 8, 1. Now answer the following questions:

a. What is the atomic number of this element?

The atomic number of this element is 11

b. What is the group of this element?

Since it has 1 valence electron, it belongs to Group 1 (Alkali metals).

c. To which period does the element belong?

Since it has 3 shells, it belongs to the third period.